Benjamin D. Sauer

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PROFESSIONAL SUMMARY

Engineering student focused on dynamics, controls, and electromechanical systems with experience in simulation, sensor integration, and hardware prototyping.

EDUCATION

Harvey Mudd College, Claremont, CA *May 2028*

- *Bachelor of Science in Engineering,* Cumulative GPA: 3.73
- Coursework: Circuit Analysis, Elastic Dynamics/Vibrations, Systems Engr, Design/Manufacturing

Pomona College, Claremont, CA *2024-2026*

- Coursework: Physics (Mechanics, Electromagnetism, Modern), Differential Equations

TECHNICAL EXPERIENCE

Ballistic Spore Dispersal Research Assistant, Pomona College *May 2025-Present*

- Simulate compressible airflow during spore ejection with ANSYS Fluent and analyze vortex flow
- Design and fabricate aluminum model of moss capsule using CAD, machining, and Arduino actuation
- Compare high-speed spore footage (MATLAB computer vision) with simulation to guide iterations

Autonomous Underwater Vehicle: Depth Control *August 2025-December 2025*

- Integrated and calibrated pressure sensor and thermistor to collect vehicle depth and temperature
- Applied PID control on PWM-controlled thrusters to reach target depth within 1.07% measured error
- Benchtop tested electromechanical system and sensors with oscilloscope and multimeter

High Speed Railcar Suspension Dynamics Modeling *December 2025-March 2026*

- Derived multi-DOF equations of motion and state-space model for Python simulation
- Analyzed mode shapes and transient response to step and stochastic track disturbances
- Designed skyhook semi-active suspension control, reducing RMS car-body acceleration by 17.4%

Compost Extractor Machine Product Development *March 2026-May 2026*

- Led 5 engineers through design of a compost extraction machine, coordinating meetings & deliverables
- Designed 7 SolidWorks subassemblies and fabricated to test power transmission and extraction rates

ADDITIONAL EXPERIENCE

Civil Engineering Intern, Fuscoe Engineering *June 2026-Aug 2026*

- Prepared AutoCAD plan sets for 8+ residential and commercial land-development projects
- Caught utility-placement conflicts and notation errors across four plan documents, preventing redesign

President & Founder, STEM Bridges, HMC Civic & Community Engagement *July 2026-Present*

- Founded a project-based engineering mentorship program inspiring curiosity in engineering among middle school students with fewer STEM opportunities

SKILLS

- Controls & Electronics: PID and skyhook control, state-space modeling, sensor integration and calibration, PWM motor control, Arduino, oscilloscope and multimeter
- CAD: SolidWorks (assemblies, motion, basic FEA), Fusion 360, AutoCAD
- Software: MATLAB, Python, Excel, ANSYS Fluent
- Fabrication: Manual lathe and mill, CNC machining, metal and wood fabrication, 3D-Printing

AWARDS AND HONORS

- Hispanic Scholarship Fund National Scholar 2022-Present and 2025 & 2026 Scholarship Recipient